BT Tuần 2.

Bài 1.

a. Ta có ma trận mở rộng:

$$\left[\begin{array}{ccc|c} 4 & -1 & 1 & 8 \\ 2 & 5 & 2 & 3 \\ 1 & 2 & 4 & 11 \end{array}\right] \xrightarrow[R_3 - \frac{1}{4}R_1 \to R_3]{R_2 - \frac{1}{2}R_1 \to R_2} \left[\begin{array}{ccc|c} 4 & -1 & 1 & 8 \\ 0 & \frac{11}{2} & \frac{3}{2} & -1 \\ 0 & \frac{9}{4} & \frac{15}{4} & 9 \end{array}\right]$$

$$\xrightarrow{R_3 - \frac{9}{22}R_2 \to R_3} \left[\begin{array}{ccc|c} 4 & -1 & 1 & 8 \\ 0 & \frac{11}{2} & \frac{3}{2} & -1 \\ 0 & 0 & 3 & 9 \end{array}\right] \xrightarrow[\substack{R_1 \cdot \frac{1}{4} \to R_1 \\ R_2 \cdot \frac{2}{11} \to R_2}]{R_3 \cdot \frac{1}{3} \to R_3} \left[\begin{array}{ccc|c} 1 & -\frac{1}{4} & \frac{1}{4} & 2 \\ 0 & 1 & \frac{3}{11} & -\frac{2}{11} \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$$\xrightarrow{R_1 + \frac{1}{4}R_2 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & \frac{14}{44} & \frac{86}{44} \\ 0 & 1 & \frac{3}{11} & -\frac{2}{11} \\ 0 & 0 & 1 & 3 \end{array}\right] \xrightarrow[R_2 - \frac{3}{11}R_3 \to R_2]{R_1 - \frac{14}{44}R_3 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

b. Ta có ma trận mở rộng

$$\left[\begin{array}{ccc|c} 4 & 1 & 2 & 9 \\ 2 & 4 & -1 & -5 \\ 1 & 1 & -3 & -9 \end{array}\right] \xrightarrow{R_1 \cdot \frac{1}{4} \to R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{4} & \frac{1}{2} & \frac{9}{4} \\ 2 & 4 & -1 & -5 \\ 1 & 1 & -3 & -9 \end{array}\right]$$

$$\xrightarrow[R_3 - R_1 \to R_3]{R_2 - 2R_1 \to R_2} \left[\begin{array}{ccc|c} 1 & \frac{1}{4} & \frac{1}{2} & \frac{9}{4} \\ 0 & \frac{7}{2} & -2 & -\frac{19}{2} \\ 0 & \frac{3}{4} & -\frac{7}{2} & -\frac{45}{4} \end{array}\right] \xrightarrow{R_2 \cdot \frac{2}{7} \to R_2} \left[\begin{array}{ccc|c} 1 & \frac{1}{4} & \frac{1}{2} & \frac{9}{4} \\ 0 & 1 & -\frac{4}{7} & -\frac{19}{7} \\ 0 & \frac{3}{4} & -\frac{7}{2} & -\frac{45}{4} \end{array}\right]$$

$$\cancel{R_3 \times \frac{-14}{43}} \xrightarrow{R_3 - 0{,}75R_2 \to R_3} \left[\begin{array}{ccc|c} 1 & \frac{1}{4} & \frac{1}{2} & \frac{9}{4} \\ 0 & 1 & -\frac{4}{7} & -\frac{19}{7} \\ 0 & 0 & -\frac{43}{14} & -\frac{129}{14} \end{array}\right] \xrightarrow{R_3 \cdot \left(-\frac{14}{43}\right)} \left[\begin{array}{ccc|c} 1 & \frac{1}{4} & \frac{1}{2} & \frac{9}{4} \\ 0 & 1 & -\frac{4}{7} & -\frac{19}{7} \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$$\xrightarrow[R_1 - 0{,}5R_3]{R_2 + \frac{4}{7}R_3} \left[\begin{array}{ccc|c} 1 & \frac{1}{4} & 0 & \frac{3}{4} \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right] \xrightarrow{R_1 - \frac{1}{4}R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Bài 2:

a. Ta có ma trận hệ số mở rộng:

$$\left[\begin{array}{ccc|c} -1 & 4 & 1 & 8 \\ \frac{5}{2} & \frac{2}{3} & \frac{2}{3} & 1 \\ 2 & 1 & 4 & 11 \end{array}\right] \xrightarrow{R_1 \times (-1) \to R_1} \left[\begin{array}{ccc|c} 1 & -4 & -1 & -8 \\ \frac{5}{2} & \frac{2}{3} & \frac{2}{3} & 1 \\ 2 & 1 & 4 & 11 \end{array}\right] \xrightarrow[R_3 - 2R_1]{R_2 - \frac{5}{2}R_1} \left[\begin{array}{ccc|c} 1 & -4 & -1 & -8 \\ 0 & \frac{22}{3} & \frac{7}{3} & \frac{43}{3} \\ 0 & 9 & 6 & 27 \end{array}\right]$$

$$\xrightarrow{R_2 \cdot \frac{3}{22} \to R_2} \left[\begin{array}{ccc|c} 1 & -4 & -1 & -8 \\ 0 & 1 & \frac{7}{22} & \frac{43}{22} \\ 0 & 9 & 6 & 27 \end{array}\right] \xrightarrow[R_3 - 9R_2]{R_1 + 4R_2} \left[\begin{array}{ccc|c} 1 & 0 & \frac{3}{11} & -\frac{2}{11} \\ 0 & 1 & \frac{7}{22} & \frac{43}{22} \\ 0 & 0 & \frac{69}{22} & \frac{207}{22} \end{array}\right] \xrightarrow{R_3 \cdot \frac{22}{69}} \left[\begin{array}{ccc|c} 1 & 0 & \frac{3}{11} & -\frac{2}{11} \\ 0 & 1 & \frac{7}{22} & \frac{43}{22} \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$$\xrightarrow[R_2 - \frac{7}{22}R_3]{R_1 - \frac{3}{11}R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

b. Ta có ma trận hệ số mở rộng

$$\left[\begin{array}{ccc|c} 4 & 2 & -1 & -5 \\ \frac{1}{9} & \frac{1}{9} & -\frac{1}{3} & -1 \\ 1 & 4 & 2 & 9 \end{array}\right] \xrightarrow{R_1 \cdot \frac{1}{4} \to R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{4} & -\frac{5}{4} \\ \frac{1}{9} & \frac{1}{9} & -\frac{1}{3} & -1 \\ 1 & 4 & 2 & 9 \end{array}\right]$$

$$\xrightarrow[R_3 - R_1]{R_2 - \frac{1}{9}R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{4} & -\frac{5}{4} \\ 0 & \frac{1}{18} & -\frac{11}{36} & -\frac{31}{36} \\ 0 & \frac{7}{2} & \frac{9}{4} & \frac{41}{4} \end{array}\right] \xrightarrow{R_2 \cdot 18 \to R_2} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{4} & -\frac{5}{4} \\ 0 & 1 & -\frac{11}{2} & -\frac{31}{36} \\ 0 & \frac{7}{2} & \frac{9}{4} & \frac{41}{4} \end{array}\right]$$

$$\xrightarrow[R_3 - \frac{7}{2}R_2]{R_1 - \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 0 & \frac{5}{2} & \frac{13}{2} \\ 0 & 1 & -\frac{11}{2} & -\frac{31}{2} \\ 0 & 0 & \frac{43}{2} & \frac{129}{2} \end{array}\right] \xrightarrow{R_3 \cdot \frac{2}{43}} \left[\begin{array}{ccc|c} 1 & 0 & \frac{5}{2} & \frac{13}{2} \\ 0 & 1 & -\frac{11}{2} & -\frac{31}{2} \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$$\xrightarrow[R_2 + \frac{11}{2}R_3]{R_1 - \frac{5}{2}R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Bài 3: a Ta có ma trận hệ số mở rộng

$$\left[\begin{array}{ccc|c} 1 & -1 & 3 & 2 \\ 3 & -3 & 1 & -1 \\ 1 & 1 & 0 & 3 \end{array}\right] \xrightarrow[R_3 - R_1]{R_2 - 3R_1} \left[\begin{array}{ccc|c} 1 & -1 & 3 & 2 \\ 0 & 0 & -8 & -7 \\ 0 & 2 & -3 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & -1 & 3 & 2 \\ 0 & 2 & -3 & 1 \\ 0 & 0 & -8 & -7 \end{array}\right] \xrightarrow[R_2 \cdot \frac{1}{2}]{R_3\left(\frac{-1}{8}\right)} \left[\begin{array}{ccc|c} 1 & -1 & 3 & 2 \\ 0 & 1 & -\frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{7}{8} \end{array}\right]$$

$$\xrightarrow[R_2 + \frac{3}{2}R_3]{R_1 - 3R_3} \left[\begin{array}{ccc|c} 1 & -1 & 0 & -\frac{5}{8} \\ 0 & 1 & 0 & \frac{29}{16} \\ 0 & 0 & 1 & \frac{7}{8} \end{array}\right] \xrightarrow{R_1 + R_2 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 19/16 \\ 0 & 1 & 0 & 29/16 \\ 0 & 0 & 1 & \frac{7}{8} \end{array}\right]$$

b
$$\left[\begin{array}{ccc|c} 2 & -1,5 & 3 & 1 \\ -1 & 0 & 2 & 3 \\ 4 & -4,5 & 5 & 1 \end{array}\right] \xrightarrow[R_3 - 2R_1]{R_2 + 0,5R_1} \left[\begin{array}{ccc|c} 2 & -1,5 & 3 & 1 \\ 0 & -0,75 & 3,5 & 3,5 \\ 0 & -1,5 & -1 & -1 \end{array}\right]$$

$$\xrightarrow{R_3 - 2R_2} \left[\begin{array}{ccc|c} 2 & -1,5 & 3 & 1 \\ 0 & -0,75 & 3,5 & 3,5 \\ 0 & 0 & -8 & -8 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

c
$$\left[\begin{array}{ccc|c} 2 & 0 & 0 & 3 \\ 1 & 1,5 & 0 & 4,5 \\ 0 & -3 & 0,5 & -6,6 \\ 2 & -2 & 1 & 0,8 \end{array}\right] \xrightarrow[R_4 - R_1]{R_2 - \frac{1}{2}R_1} \left[\begin{array}{ccc|c} 2 & 0 & 0 & 3 \\ 0 & 1,5 & 0 & 3 \\ 0 & -3 & 0,5 & -6,6 \\ 0 & -2 & 1 & -2,2 \end{array}\right]$$

$$\xrightarrow[R_4 + \frac{4}{3}R_2]{R_3 + 2R_2} \left[\begin{array}{ccc|c} 2 & 0 & 0 & 3 \\ 0 & 1,5 & 0 & 3 \\ 0 & 0 & 0,5 & -0,6 \\ 0 & 0 & 1 & 1,8 \end{array}\right] \xrightarrow{R_4 - 2R_3} \left[\begin{array}{ccc|c} 2 & 0 & 0 & 3 \\ 0 & 1,5 & 0 & 3 \\ 0 & 0 & 0,5 & -0,3 \\ 0 & 0 & 0 & 3 \end{array}\right]$$

$\Rightarrow$ hệ pt vô n°.

d

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 2 \\ 2 & 1 & -1 & 1 & 1 \\ 4 & -1 & -2 & 2 & 0 \\ 3 & -1 & -1 & 2 & -3 \end{array}\right] \xrightarrow[\substack{R_3-4R_1 \\ R_4-3R_1}]{R_2-2R_1} \left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 2 \\ 0 & -1 & -1 & -1 & -3 \\ 0 & -5 & -2 & -2 & -8 \\ 0 & -4 & -1 & -1 & -9 \end{array}\right]$$

$$\xrightarrow[R_4-4R_2]{R_3-5R_2} \left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 2 \\ 0 & -1 & -1 & -1 & -3 \\ 0 & 0 & 3 & 3 & 7 \\ 0 & 0 & 3 & 3 & 3 \end{array}\right] \xrightarrow{R_4-R_3} \left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 2 \\ 0 & -1 & -1 & -1 & -3 \\ 0 & 0 & 3 & 3 & 7 \\ 0 & 0 & 0 & 0 & -4 \end{array}\right]$$

$\Rightarrow$ hpt vô nghiệm

Bài 4:

a. Ta có sơ đồ Horner.

+	7	-8	0	7	18	-9	-20
2		14	12	24	62	160	302
	7	6	12	31	80	151	282

$\Rightarrow$ Số dư $b_0 = 282$.

$Q(x) = 7x^5 + 6x^4 + 12x^3 + 80x + 151$

$\Rightarrow P(x) = (x-2)\,Q(x) + 282$

b

−	7	-8	0	7	18	-9	-20	
2		~~14~~	~~12~~	~~24~~	~~62~~	~~160~~	~~302~~	
2	7	-22	16	7	4	-45	-2	40

$\Rightarrow P(x)(x-2) = 7x^7 - 22x^6 + 16x^5 + 7x^4 + 4x^3 - 45x^2 - 2x + 40$

$P(x)\,(x-3)$

−	7	-8	0	7	18	-9	-20	
3	7	-29	24	7	-3	-63	7	60

$= 7x^7 - 29x^6 + 24x^5 + 7x^4 - 3x^3 - 63x^2 + 7x + 60$

Bài 6:

$$R(x) = 1 + x(3 + (x-1)(4 + (x-2)(-7 + (x-3)5)))$$

$R_3(x) = 5(x-3) - 7 = 5x - 22.$

$$\begin{array}{c|cccc} -0 & 5 & -22 & \\ 2 & 5 & -32 & 44 \\ \hline & \cancel{5} & \cancel{-32} & \end{array}$$

$R_2(x) = 4 + (x-2)(-7 + 5(x-3))$

$= 5x^2 - 32x + 48$

$$\begin{array}{c|cccc} - & 5 & -32 & 48 & \\ 1 & 5 & -37 & 80 & -48. \end{array}$$

$R_1(x) = 3 + (x-1)R_2(x)$

$= 5x^3 - 37x^2 + 80x - 45$

$R(x) = 1 + x.R_1(x).$

$= 5x^4 - 37x^3 + 80x^2 - 45x + 1.$

Bài 5:

$w(x) = (x-2)(x-4)(x-5)(x-7)$

$R_1(x) = (x-2)(x-4).$

$= x^2 - 6x + 8$

$$\begin{array}{c|ccc} - & 1 & -2 & \\ 4 & 1 & -6 & 8. \end{array}$$

$R_2(x) = R_1(x)(x-5)$

$$\begin{array}{c|ccc} - & 1 & -6 & 8 \\ 5 & 1 & -1 & 3 \end{array}$$

$= x^3 - x^2 + 3x - 15$

$$\begin{array}{c|cccc} - & 1 & -1 & 3 & -15 \\ 7 & 1 & 6 & 45 & 300 \end{array}$$

$w(x) = R_2(x)(x-7)$

$= \cancel{x^4 - 18x^3 + 117x^2 - 330x + 280}$

$= x^4 + 6x^3 + 45x^2 + 300x + 280$